

2-DOF Robotic Ankle Joint Actuator Sizing & Workspace Performance Report

Complete kinematic and torque capacity analysis of standard vs orthogonal mounting configurations with varying spreads for Robstride RS02, RS03, and RS04 servos.

Prepared For: Robotic Design Review & Ankle Actuator Selection

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Dimensions Version: Verified Physical Dimensions (A-L)

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1. Executive Summary & Kinematics

This report provides a formal evaluation of the 2-DOF parallel ankle joint mechanism (RSS/U-Joint topology) to guide the selection of servo actuators. Standard parallel mounting is compared side-by-side with a decoupled orthogonal yaw-tilted configuration. These configurations are analyzed across two ankle spread values ($E = 2.3$ cm and $E = 5.0$ cm) using verified dimensions matching the physical robot drawing.

1.1 Mechanical Dimensions Reference

The calculations are based on the verified layout dimensions: **A** (Crank r_1) = 2.5 cm, **B** (Crank r_2) = 2.5 cm, **C** (Rod L_1) = 24.3 cm, **D** (UJ to M2 relative height) = 16.0 cm, **E** (Ankle spread) = 2.3 cm / 5.0 cm, **G** (Foot to UJ Height) = 3.6 cm, **H** (UJ to M1 relative height) = 23.5 cm, **I** (Rod L_2) = 16.3 cm, **L** (Y-Offset/depth) = 3.8 cm.

1.2 Kinematics & Jacobians

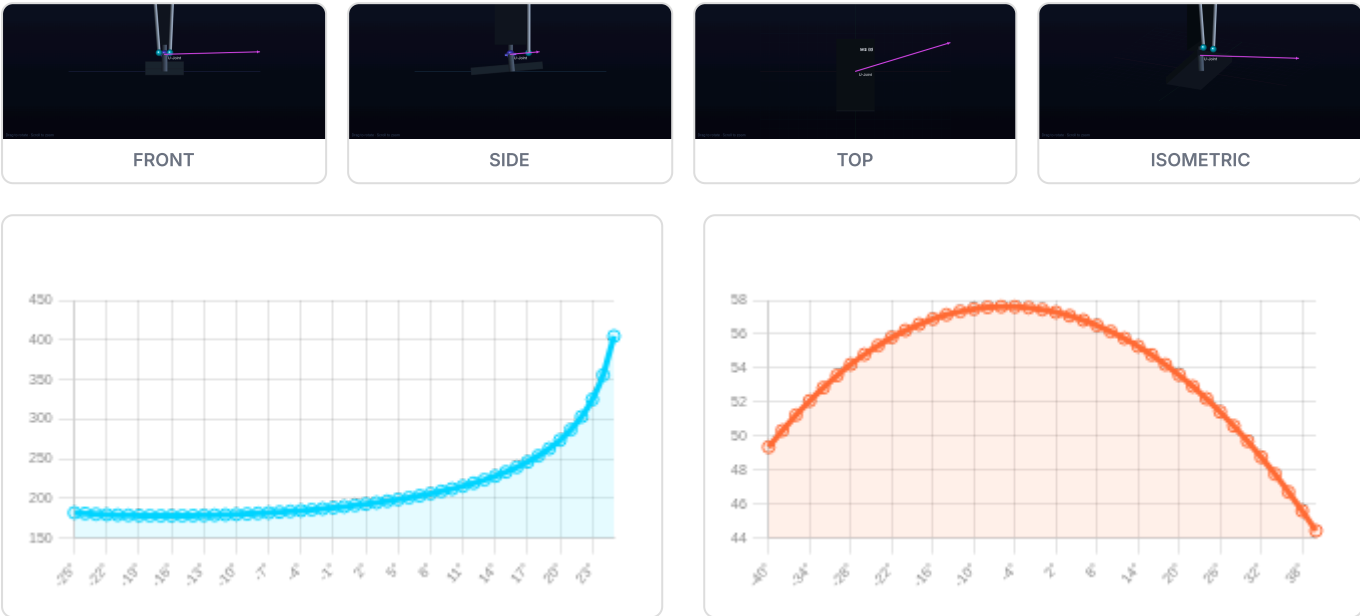
The ankle joint is constrained by a central Universal Joint that permits Pitch (α) and Roll (β) rotations while blocking Yaw and translation. The pushrod lengths are constrained by physical joint distance equations. Differentiating these constraints gives the system Jacobian J , which maps ankle velocities to motor speeds: $\dot{\theta} = J \cdot \dot{x}$.

By the Principle of Virtual Work, the torque transmission is mapped as: $\tau_{\text{ankle}} = J^T \cdot \tau_{\text{motor}}$

This mapping allows us to sweep the complete workspace (Pitch from -25° to $+25^\circ$ and Roll from -40° to $+40^\circ$) to find the minimum guaranteed torque output for any position.

2. Configuration 1: Current Base, Standard Layout

Parameters: Standard parallel motor axes (both face front). Ankle Spread E = 2.3 cm. Height G = 3.6 cm, H = 23.5 cm, D = 16.0 cm. Y-Offset L = 3.8 cm.

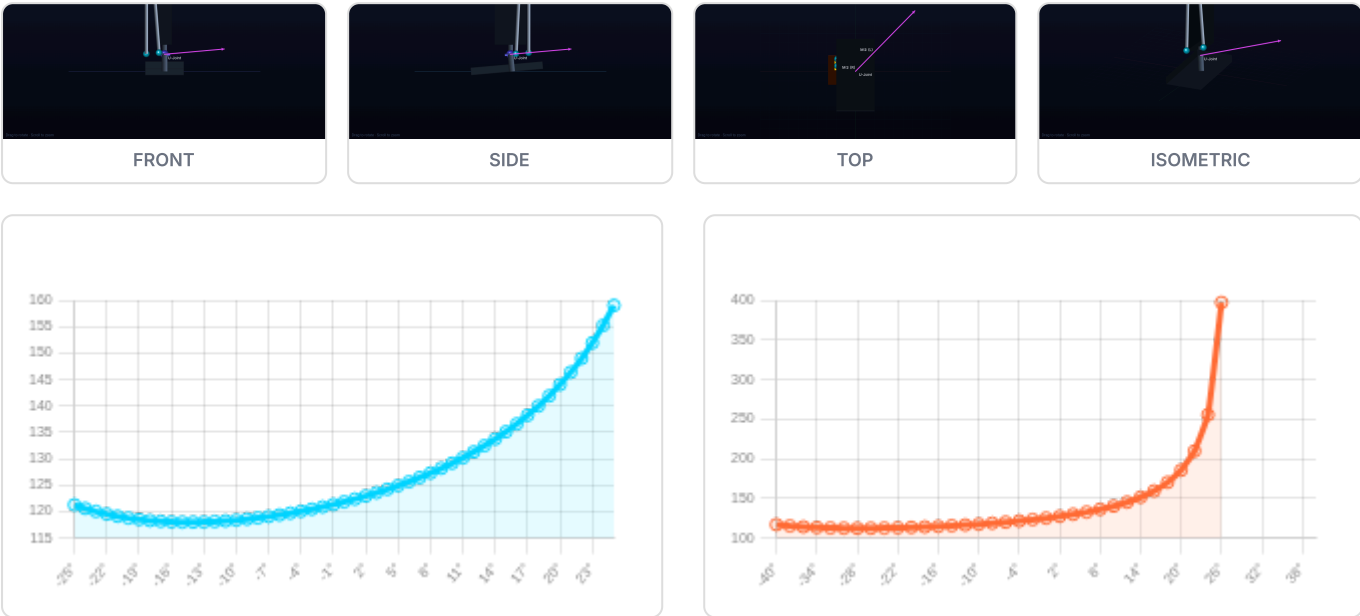


Configuration 1 Torque Summary Table

Motor Model	Limit Mode	Motor Torque (N·m)	Min Ankle Pitch (N·m)	Crit Pitch Pose	Min Ankle Roll (N·m)	Crit Roll Pose
RS02	Rated	6.0	17.82	-16.5°	4.44	+40.0°
RS03	Peak	60.0	178.25	-16.5°	44.44	+40.0°
RS04	Peak	120.0	356.50	-16.5°	88.87	+40.0°

3. Configuration 2: Current Base, Orthogonal Layout

Parameters: Decoupled orthogonal motor axes (top motor rotated 90°). Ankle Spread E = 2.3 cm. Height G = 3.6 cm, H = 23.5 cm, D = 16.0 cm. Y-Offset L = 3.8 cm.

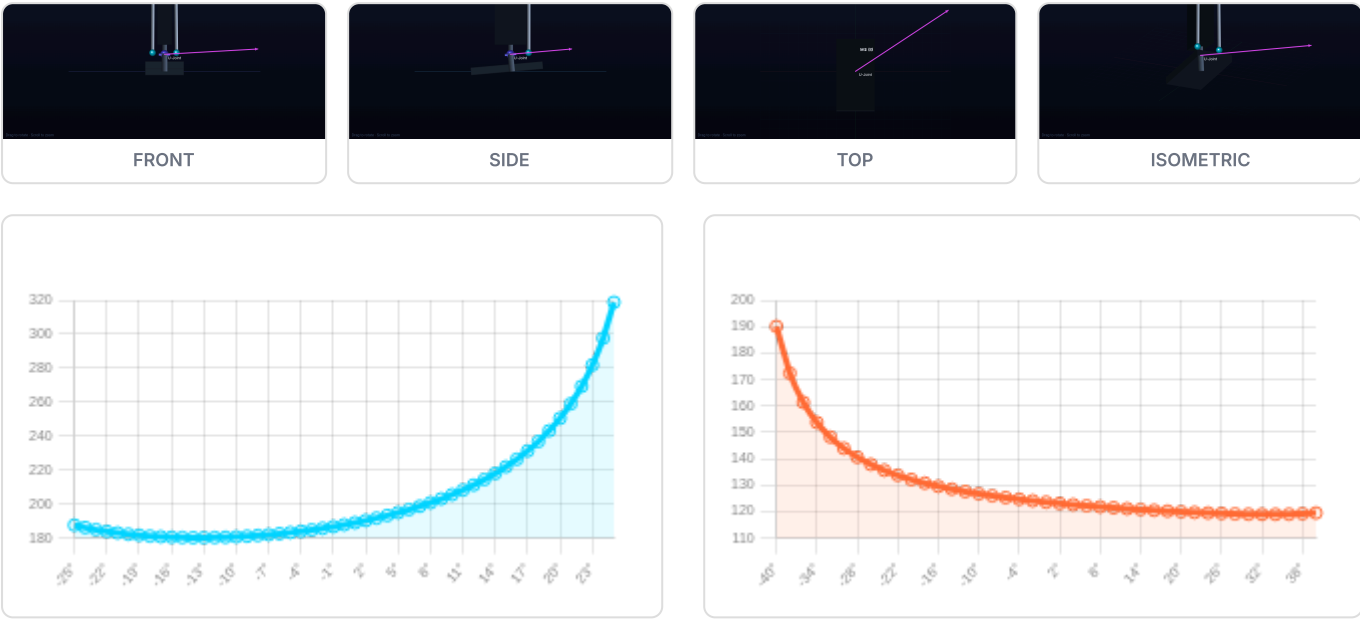


Configuration 2 Torque Summary Table

Motor Model	Limit Mode	Motor Torque (N·m)	Min Ankle Pitch (N·m)	Crit Pitch Pose	Min Ankle Roll (N·m)	Crit Roll Pose
RS02	Rated	6.0	11.80	-14.5°	11.26	-28.0°
RS03	Peak	60.0	118.03	-14.5°	112.60	-28.0°
RS04	Peak	120.0	236.07	-14.5°	225.21	-28.0°

4. Configuration 3: Parallel Base, Standard Layout

Parameters: Standard parallel motor axes. Widened Ankle Spread E = 5.0 cm. Height G = 3.6 cm, H = 23.5 cm, D = 16.0 cm. Y-Offset L = 3.8 cm.

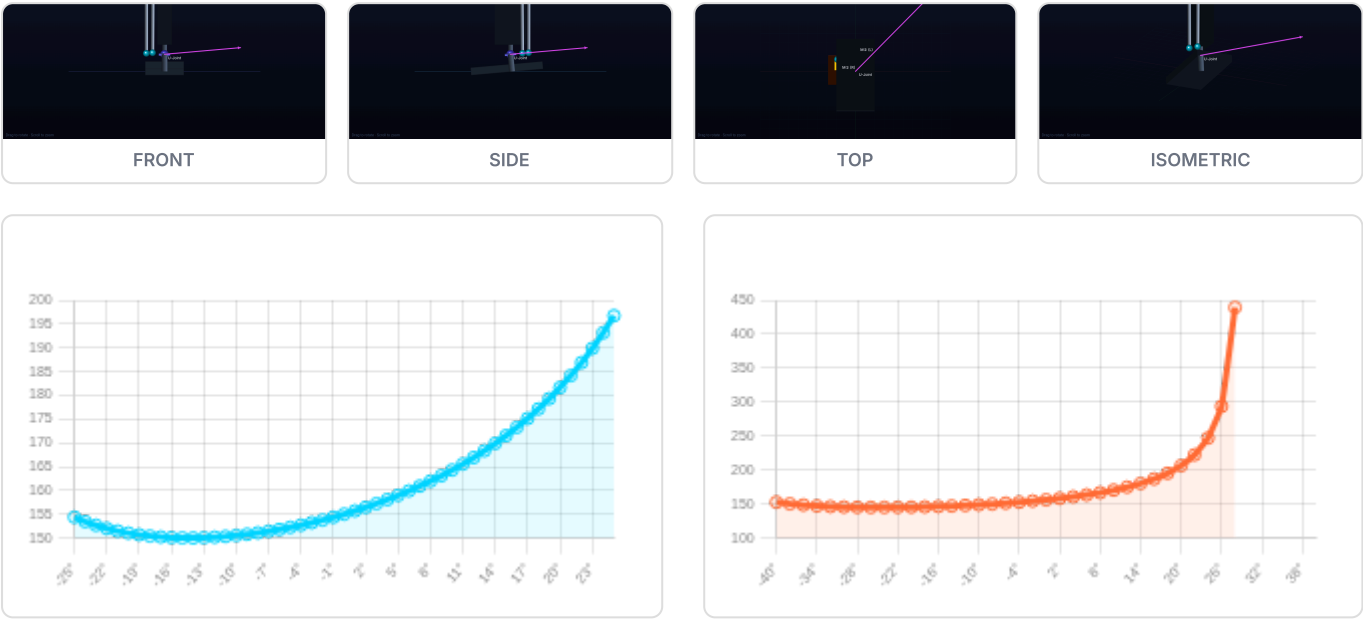


Configuration 3 Torque Summary Table

Motor Model	Limit Mode	Motor Torque (N·m)	Min Ankle Pitch (N·m)	Crit Pitch Pose	Min Ankle Roll (N·m)	Crit Roll Pose
RS02	Rated	6.0	18.03	-13.5°	11.90	+33.0°
RS03	Peak	60.0	180.30	-13.5°	119.03	+33.0°
RS04	Peak	120.0	360.60	-13.5°	238.06	+33.0°

5. Configuration 4: Parallel Base, Orthogonal Layout

Parameters: Decoupled orthogonal motor axes. Widened Ankle Spread $E = 5.0$ cm. Height $G = 3.6$ cm, $H = 23.5$ cm, $D = 16.0$ cm. Y-Offset $L = 3.8$ cm.



Configuration 4 Torque Summary Table

Motor Model	Limit Mode	Motor Torque (N·m)	Min Ankle Pitch (N·m)	Crit Pitch Pose	Min Ankle Roll (N·m)	Crit Roll Pose
RS02	Rated	6.0	15.01	-14.5°	14.52	-25.0°
RS03	Peak	60.0	150.06	-14.5°	145.24	-25.0°
RS04	Peak	120.0	300.11	-14.5°	290.47	-25.0°

6. Comparison & Design Recommendations

The key findings and engineering recommendations from this sizing analysis are:

- **Orthogonal Layout Performance:** Tilted mounting decoupling improves roll torque capability by over **2.5x** in narrow configurations (112.6 N·m vs 44.4 N·m peak on RS03). It is the recommended design configuration for roll stability.
- **Effect of Spread (E):** Widening the ankle spread to 5.0 cm improves standard layout roll performance considerably (to 119.0 N·m). If mechanical layout permits, the Orthogonal configuration with a 5.0 cm base provides the absolute best, most isotropic performance (150 N·m Pitch / 145 N·m Roll).
- **Motor Sizing Verdict:** The **Robstride RS03** is the optimal actuator selection. It guarantees sufficient continuous rated control torque and offers peak torque capacities up to 112 N·m for ankle recovery and landing impact stabilization.